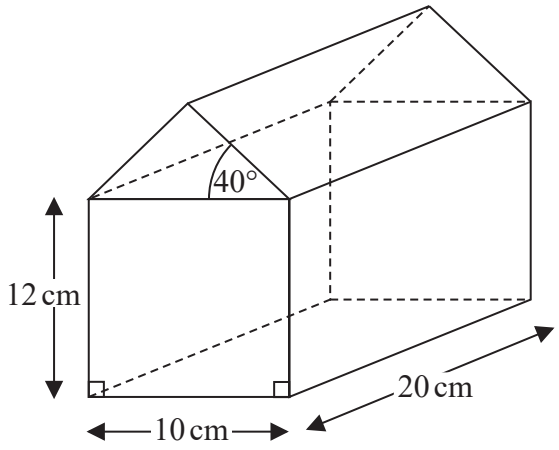
**9** The diagram shows a prism.




The cross section of the prism has exactly one line of symmetry.

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

Symmetric roof: half-width = 10/2 = 5 cm.

Roof height = 5 tan 40° ≈ 4.1955 cm.

Cross-section area = 10 × 12 + ½ × 10 × 4.1955…

≈ 140.9775 cm². Volume = area × 20

≈ 2819.55… cm³ = 2820 cm³ (3 s.f.).

**5 marks total**
Half-width 5
Height 5tan40°
Rectangle + triangle
Multiply by 20
2820 cm³

..... cm³

**(Total for Question 9 is 5 marks)**

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